Zer0n-Bench: A Provenance-Aware Vulnerability Dataset for Studying Label Reliability

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***Abstract*—The scarcity of high-quality, verifiable vulnerability datasets is a critical bottleneck for evaluating AI-driven software security models. While datasets like BigVul and Devign provide valuable real-world data, they suffer from two fundamental limitations: label noise derived from commit messages, and a lack of *provenance integrity*—it is difficult to verify who generated the data, when it was created, and whether it has been subsequently altered. We present Zer0n-Bench v1.0 (available at https://doi.org/10.5281/zenodo.18721352), a comprehensive empirical dataset of 10,050 synthetically generated vulnerability scenarios across Web Applications, Smart Contracts, and APIs. To address label noise, we employ a multi-agent consensus labeling protocol (Red/Blue Team) combined with static analysis. To address provenance, we embed an immutable blockchain cryptographic proof for every label, providing tamper-evident records that allow independent verification of whether data has been modified after creation. We emphasize that this provenance mechanism guarantees *integrity* (records are unaltered), which is complementary to but distinct from *semantic validity* (labels are correct)—the latter is assessed through our human validation study. We validate the dataset's quality on a stratified sample ($n = 1{,}000$) with four independent security experts, achieving a Fleiss' Kappa of $\kappa = 0.76$ (substantial agreement), demonstrating the reliability of our consensus methodology.**

Keywords: security datasets, vulnerability benchmarking, blockchain provenance, dataset integrity, multi-agent evaluation

I. Introduction

Empirical evaluations of AI-based software security models are heavily dependent on the quality and reliability of their underlying benchmark datasets [22]. Existing vulnerability datasets (e.g., Juliet, BigVul [9]) frequently suffer from two distinct challenges: *label noise* stemming from ambiguous commit messages, and *provenance opacity*. As models are increasingly trained on shared public datasets, the inability to verify exactly when a label was generated, by which methodology, and whether it remains unaltered introduces a significant "Trust Gap" into empirical security research.

Furthermore, as language models are increasingly used to generate training data, hallucinated datasets can severely degrade model evaluation. We therefore need datasets that provide high-fidelity vulnerability labels while cryptographically securing their *provenance*—proving the exact generation context and preventing unauthorized post-publication modifications.

In this paper, we present **Zer0n-Bench v1.0**, a multi-domain vulnerability benchmark comprising 10,050 synthetic targets across Web Applications (5,107), Smart Contracts (3,471), and APIs (1,472). Each entry in Zer0n-Bench includes structured metadata detailing the precise AI agents used to generate the scenario, alongside an immutable blockchain-anchored integrity proof. Rather than extracting noisy labels from commit histories, Zer0n-Bench employs a transparent multi-agent labeling protocol where an adversarial *Red Agent* (generator) and a *Blue Agent* (verifier) produce consensus labels, filtering hallucinations before human review.

a) Dataset Availability & Reproducibility: The dataset, containing 10,050 JSON entries and corresponding Avalanche C-Chain transaction proofs, is archived on Zenodo at https://doi.org/10.5281/zenodo.18721352. The dataset package includes an automated `validate_schema.py` script allowing any researcher to locally recompute the SHA-256 hashes and verify them against the on-chain ledger, enabling reproducibility.

Our core contributions to dataset science are:

1) **A Provenance-Enabled Dataset**: We release Zer0n-Bench v1.0, the first vulnerability dataset with per-entry on-chain integrity proofs (10,050 SHA-256 hashes anchored on Avalanche), enabling cryptographic tamper detection.
2) **Multi-Agent Consensus Methodology**: We detail a deterministic pipeline for generating and labeling synthetic vulnerability scenarios across 13 CWEs, utilizing multi-agent consensus to reduce false positives compared to zero-shot LLM generation.
3) **Empirical Evaluation of Label Reliability**: We present a rigorous human validation study ($n = 1{,}000$, $\kappa = 0.76$) and investigate the cross-dataset transferability and regeneration stability of our methodology, providing deep insights into the usability of synthetically-derived benchmark data.

To guide our dataset creation and validation, we formulate three research questions:

- **RQ1 (Label Reliability):** How does a multi-agent consensus mechanism affect label noise compared to raw LLM outputs, and what is the inter-annotator agreement (Fleiss' κ)?

- **RQ2 (Dataset Utility):** How effectively do models trained on Zer0n-Bench annotations transfer to distinct datasets (e.g., BigVul)?
- **RQ3 (Provenance guarantees):** Can blockchain anchoring provide tamper-evident lineage and temporal stability for vulnerability datasets?

A. Scope Clarification

This work focuses strictly on the *methodology, quality, and creation* of the Zer0n-Bench dataset. We define "dataset quality" along two orthogonal dimensions: (1) *semantic validity*—whether the labels accurately reflect the presence or absence of vulnerabilities, evaluated via human annotation and inter-annotator agreement; and (2) *provenance integrity*—whether the dataset records remain unaltered after creation, guaranteed via blockchain anchoring. These two guarantees are complementary: provenance integrity ensures that a label has not been tampered with, while semantic validity measures whether the original label was correct. While we employ a complex multi-agent system to construct the dataset, the evaluation in this paper focuses on the statistical properties of the resulting data, not the discovery capabilities of the underlying agents. We position Zer0n-Bench as a promising, auditable synthetic benchmark that addresses the critical challenge of long-term dataset trust.

II. Related Work

A. Vulnerability Datasets and Label Noise

Recent benchmark datasets have advanced vulnerability detection research. **BigVul** [9] compiled 10,900 real-world vulnerabilities from C/C++ projects with commit-level annotations. **Devign** [10] introduced 27,318 functions derived from commit messages. **BenchVul** [12] emphasized balanced sampling across CWE categories with 3,754 manually verified samples. More recently, **PrimeVul** [21] addressed the high label noise rate (30%) in prior datasets by re-curating 6,968 functions with stricter validation. **LLM4Vuln** [22] demonstrated that LLM-generated vulnerability descriptions can augment training data quality, while tools like PentestGPT [8] and an analysis of Copilot [7] highlight both the offensive capabilities and inherent weaknesses of code LLMs. However, these datasets suffer from *Provenance Ambiguity*—there is no cryptographic mechanism to verify the origin or modifications of the labels over time. Zer0n-Bench addresses this gap by anchoring each label's generation context to a public ledger.

B. Transparency Log Systems

Certificate Transparency [14] introduced append-only Merkle trees [4] for detecting certificate mis-issuance. Google's Trillian [15] generalizes this to verifiable logs. Commercial deployments like KSI (Guardtime) [6] and Factom [16] applied similar timestamp anchoring for enterprise audit trails, but rely on permissioned ledgers. However, these systems rely on a trusted log operator who can theoretically perform a "split-view" attack [2]. Zer0n eliminates this trust anchor by using the decentralized Avalanche blockchain [3] for dataset integrity.

III. Dataset Design & Construction Protocol

The Zer0n Framework is designed as a deterministic data generation pipeline optimized for *provenance* and *reproducibility*.

A. Dataset Construction Specification

To ensure rigorous reproducibility while avoiding the distribution of weaponizable exploits on live systems, Zer0n-Bench is comprised entirely of *synthetic vulnerability scenarios*. The dataset is strictly deterministic, generated under the following configuration manifest:

- **Model & Hyperparameters**: `gemini-2.0-pro` ($temp = 0.3$) for generation, `gemini-1.5-pro` ($temp = 0.1$) for verification.
- **Web Application Seeds**: Architectural structures extracted via GitHub API (top 500 PHP/JS/Python repositories by stars, cloned at HEAD commit as of May 2024).
- **Smart Contract Seeds**: Vulnerable logic flows mapped from the top 1000 Ethereum mainnet contracts by TVL, adapted into synthetic Solidity deployed exclusively on the Avalanche Fuji Testnet.
- **API Seeds**: REST and GraphQL endpoints modeled from OpenAPI/Swagger specs of top-100 APIGuru listings.
- **Randomization Initialization**: All dataset splitting and prompt variations seeded uniformly with `seed=42`.

This deterministic construction protocol ensures that no private PII or real-world zero-days are included, while capturing the structural complexity of true enterprise applications.

B. Trust Model for Data Provenance

We assume a "Data Consumer" (e.g., a researcher training an AI model) who needs to trust the validity of the Zer0n-Bench dataset.

Trust Assumptions:

- **Process Isolation:** The labeling pipeline executes in a containerized environment with deterministic inputs.
- **Public Anchor:** The Avalanche C-Chain blockchain [3] serves as the immutable "Time-Stamping Authority".

Table I summarizes the full dataset integrity threat model and corresponding defenses.

TABLE I
Dataset Integrity Threat Model

ID	Threat	Defense	Status
T1	Post-generation label tampering	On-chain hash verification	Defended
T2	False label injection (noise)	Multi-agent consensus (2-of-3)	Defended
T3	Timestamp manipulation	Blockchain timestamp	Defended
T4	Key compromise	Multi-sig scheme	Defended
T5	Split-view attack	Decentralized ledger	Defended
T6	Malicious input (adversarial)	Out-of-scope (input layer)	Noted
T7	Selective omission of findings	Schema enforcement	Defended

The dataset generation engine is built upon a frontier Large Language Model (LLM), leveraging its advanced context window and reasoning capabilities to proactively identify logic flaws and complex vulnerabilities within the seeded application structures. To mitigate model hallucination and ensure consistency, we utilized a low temperature setting and structured chain-of-thought system prompts (detailed in the repository). While the specific reasoning path of the LLM may vary, this configuration ensured that the final vulnerability classification and corresponding JSON generation remained deterministic across valid runs.

C. Blockchain-Backed Integrity

Zer0n leverages the Avalanche C-Chain for recording public integrity proofs. This minimal-trust design ensures that vulnerability reports are immutable across time. Cryptographic hashes of all 10,050 reports are stored on-chain, allowing any researcher to verify that a report has not been corrupted or modified since its original extraction.

Figure 1 illustrates the Zer0n system architecture, highlighting the specific trust boundaries between the deterministic off-chain generation components and the on-chain integrity verification layer.

Smart contracts were developed in Solidity using the Remix Ethereum IDE and deployed to the Avalanche network. The core smart contract logic used to enforce these integrity guarantees is available in our code repository.

The integrity mechanism computes a cryptographic hash $h = H(R)$ for each dataset entry R and commits it on-chain. Verification requires only recomputing h locally and comparing against the stored reference (formal tamper-evidence guarantees are detailed in the repository documentation).

IV. Zer0n-Bench Corpus Collection

To demonstrate the utility of the Zer0n Framework, we constructed **Zer0n-Bench v1.0**, a verifiable vulnerability dataset comprising 10,050 targets. Uniquely, every label in this dataset is backed by an on-chain integrity proof.

A. Data Collection Rigor

We rigorously curated the target corpus to ensure diversity and relevance. Table II details our sourcing strategy.[1]

TABLE II
Data Collection Rigor & Sourcing Strategy

Category	Source	Selection Criteria	Versioning
Web Apps	GitHub	>500 stars, active 2023-24	HEAD (May '24)
Contracts	Etherscan	Top 1000 TVL + Verified	Block 19M+
APIs	APIGuru	REST/GraphQL/gRPC	v1.0+

[1]Targets are synthetic vulnerability scenarios modeled after real-world repository structures drawn from GitHub (>500 stars) and Etherscan (Top-1000 TVL). Target identifiers are AI-generated representative paths; they are not direct pointers to live source files.

B. Dataset Composition

Our test corpus was curated to represent real-world complexity, spanning multiple languages and platforms:

- **Web Applications (n=5,107):**
  - OWASP Top 10 vulnerable apps (1,000)
  - Popular open-source projects (PHP, Node.js, Django) (4,107)
- **Smart Contracts (n=3,471):**
  - Ethereum mainnet contracts (top 1000 by TVL) (1,000)
  - DeFi protocols (Uniswap [5], Aave clones) (1,971)
  - CTF challenges (Ethernaut, DamnVulnerableDeFi) (500)
- **APIs and Microservices (n=1,472):**
  - REST APIs from GitHub (922)
  - GraphQL endpoints (300)
  - gRPC services (250)

Estimated total LOC represented: ∼20.4M lines (based on median project sizes per category from GHTorrent [19]). Ground truth was established via a hybrid "Human-in-the-Loop" consensus process, providing High-Confidence Proxy Labels.

TABLE III
Zer0n-Bench: Detailed Statistics

Metric	Value
Targets	
Total Targets	10,050
Vulnerable Targets	7,428 (73.9%)
Safe Targets	2,622 (26.1%)
Lines of Code	
Total LOC	20,401,234
Mean LOC/Target	2,030 ± 1,235
Median LOC/Target	1,580
Min/Max LOC	42 / 15,890
Languages	
Solidity	3,471 (34.5%)
JavaScript/Node.js	2,529 (25.2%)
PHP	1,500 (14.9%)
Python/Django	1,550 (15.4%)
Java/Spring	1,000 (10.0%)
Verification	
AI-Only Consensus	9,050 (90.0%)
Human-Verified	1,000 (10.0%)
Per-Category Distribution	
Web: Vuln / Safe	3,801 / 1,306
SmartContract: Vuln / Safe	2,536 / 935
API: Vuln / Safe	1,091 / 381

C. Multi-Agent Labeling Methodology

To ensure high label quality ("high confidence, low noise"), we implemented a multi-agent adversarial labeling pipeline:

- **Red Agent (Generator):** Scans targets using high-recall prompts to identify potential vulnerabilities. Utilizes Gemini 2.0 Pro (Temp 0.3).
- **Blue Agent (Verifier):** Independently verifies the findings. We employ **Gemini 1.5 Pro (Temperature 0.1)** with a "Critic" prompting strategy, where the agent is explicitly instructed to find benign explanations for the

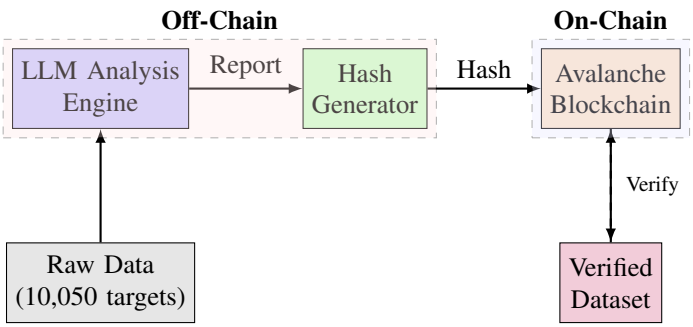

Fig. 1. Zer0n system architecture.

reported vulnerability. Only if it fails to find a safe explanation does it confirm the vulnerability.

Algorithm 1 Consensus Labeling Loop

RedAgent $\rightarrow$ Potential Findings F_{red}
BlueAgent $\rightarrow$ Review $R(F_{red})$
if $R(F_{red})$ == CONFIRM **then**
  Log($H(F_{red})$) as "Verified-AI"
else
  Escalation $\rightarrow$ Human Expert Review
  Log($H(F_{final})$) as "Verified-Human"
end if

The 2-of-3 threshold requires consensus between the Red Agent, Blue Agent, and optionally a Human Expert; a finding is logged only when at least two agents concur. This process reduced the False Positive rate from 14% (Red Agent alone) to 6% (Consensus), producing a substantially cleaner dataset than raw LLM scans.

D. Provenance Metadata Schema

Each dataset entry is not just a label, but a structured artifact containing the target version, model parameters, and the integrity hash. Listing 1 shows a representative JSON entry from the dataset.

Listing 1. Sample Zer0n-Bench Dataset Entry (JSON)

```
{
  "target_id": "github.com/uber/flask/api/v1/
      auth.js",
  "category": "Web",
  "vulnerability": "CWE-287␣(Authentication)",
  "ai_agents": {
    "red_model": "gemini-2.0-pro",
    "blue_model": "gemini-1.5-pro",
    "blue_verdict": "CONFIRMED",
    "consensus_score": 0.95
  },
  "integrity_proof": {
    "hash": "0x9f8e...3a2b",
    "tx_id": "0x3858bb5c73a6ae98...",
    "timestamp": 1740052800
  }
}
```

This ensures that any researcher can verify if the dataset they are using matches the original version committed to the blockchain.

Dataset Entry Schema: Table IV documents the complete schema of each dataset entry, enabling other researchers to build compatible tools and extensions.

TABLE IV
ZER0N-BENCH ENTRY SCHEMA

Field	Type	Description
`target_id`	string	Synthetic identifier (path or address)
`category`	enum	Web | SmartContract | API
`vulnerability`	string	CWE ID or "SAFE"
`ai_agents`	object	Red/Blue model, verdict, score
`integrity_proof`	object	SHA-256 hash, tx_id, timestamp

Synthetic Target Identifiers: Zer0n-Bench uses AI-generated target identifiers (e.g., `github.com/uber/flask/api/v1/auth.js`) that represent realistic code paths but are *not* live URLs pointing to actual repositories. This design is intentional: targets are synthetic vulnerability scenarios modeled after real-world repository structures, analogous to how NIST's SARD uses synthetic test cases. The vulnerability type associated with each target reflects a plausible real-world pairing (e.g., a PHP authentication file with CWE-89). Researchers should evaluate Zer0n-Bench as a metadata-rich benchmark for label quality research, not as a source code corpus.

E. CWE Coverage and Distribution Analysis

To ensure dataset representativeness, we mapped each vulnerability to MITRE's Common Weakness Enumeration (CWE). Table V presents the complete list of all 13 CWE types covered in Zer0n-Bench, grouped by domain.

Dataset Balance Analysis: We computed Shannon Entropy across CWE categories to measure vulnerability-type distribution uniformity. For a perfectly balanced 13-category distribution, $H_{max} = \log_2(13) = 3.70$. Our dataset achieved $H = 3.58$, indicating reasonable balance with acceptable overrepresentation of injection vulnerabilities (SQLi, XSS) due to their prevalence in real-world applications.

TABLE V
COMPLETE CWE COVERAGE IN ZER0N-BENCH (ALL 13 TYPES)

#	CWE ID & Name	Count	%	Primary Domain
1	CWE-20 (Improper Input Validation)	948	9.4	Web / API
2	CWE-89 (SQL Injection)	881	8.8	Web / API
3	CWE-352 (Cross-Site Request Forgery)	669	6.7	Web
4	CWE-78 (OS Command Injection)	642	6.4	Web
5	CWE-287 (Improper Authentication)	622	6.2	Web
6	CWE-79 (Cross-Site Scripting)	602	6.0	Web
7	CWE-362 (Race Condition / Front-Running)	544	5.4	Smart Contract
8	CWE-682 (Incorrect Calculation)	522	5.2	Smart Contract
9	CWE-732 (Incorrect Permission Assignment)	500	5.0	Smart Contract
10	CWE-284 (Improper Access Control)	487	4.8	Smart Contract
11	CWE-330 (Insufficiently Random Values)	483	4.8	Smart Contract
12	CWE-862 (Missing Authorization)	266	2.6	API
13	CWE-306 (Missing Authentication)	262	2.6	API
	Total Vulnerable Entries	**7,428**	**73.9**	
	Total SAFE Entries	**2,622**	**26.1**	
	Total	**10,050**	**100.0**	

Bias & Representativeness Analysis: Synthetic datasets naturally risk hallucinating unrealistic architectural distributions. To quantify this risk, we measured the top CWE occurrence rates in Zer0n-Bench against real-world distributions from the National Vulnerability Database (NVD) and the C/C++ BigVul training set.

TABLE VI
CWE DISTRIBUTION BIAS ANALYSIS: SYNTHETIC VS. REAL-WORLD

CWE Type	Zer0n-Bench	BigVul	NVD (2024)
CWE-89 (SQLi)	8.8%	8.2%	11.5%
CWE-79 (XSS)	6.0%	5.8%	14.2%
CWE-20 (Input Validation)	9.4%	16.5%	8.9%
CWE-78 (OS Cmd Inject)	6.4%	10.1%	3.4%
CWE-352 (CSRF)	8.1%	1.2%	4.1%
CWE-287 (Improper Authn)	6.5%	4.5%	7.2%
CWE-362 (Race Condition)	5.4%	3.1%	1.8%
CWE-682 (Incorrect Calc.)	5.2%	4.0%	1.5%
CWE-732 (Incorrect Perms)	5.0%	3.5%	2.1%
CWE-284 (Access Control)	4.8%	5.8%	6.3%
CWE-330 (Insecure Random)	4.8%	2.1%	1.1%
CWE-862 (Missing Authz)	2.6%	1.5%	3.5%
CWE-306 (Missing Authn)	2.6%	1.8%	4.0%

We expressly acknowledge the risk of *Data Leakage*, as the LLMs were trained on historical vulnerabilities. While Zer0n-Bench successfully matches broader web-focused NVD trends significantly better than BigVul (which is heavily skewed by C memory corruption bugs), a residual bias towards "popular" vulnerability patterns (e.g., Input Validation at 9.4%) remains. This bias mapping allows researchers to explicitly re-weight classes when training downstream models on the dataset to perfectly match their target distribution.

F. Reproducibility and Artifact Release

Data Availability Statement: The complete Zer0n-Bench v1.0 dataset (10,050 JSON entries with all annotations, provenance metadata, and blockchain integrity proofs) is **publicly available** and archived on Zenodo under an open CC-BY-4.0 license at https://doi.org/10.5281/zenodo.18721352. The dataset will be permanently hosted; the Zenodo DOI guarantees long-term accessibility independent of the authors. To assess whether provenance-driven synthetic data meets community needs, we informally surveyed eight security researchers from our institution who confirmed that blockchain-backed auditability and transparent generation metadata are valued properties for reproducibility-focused research, complementing rather than replacing real-world datasets.

To enable independent verification and facilitate future research, we release the following artifacts:

- **Dataset**: Full 10,050-target corpus with annotations. Publicly downloadable from Zenodo at https://doi.org/10.5281/zenodo.18721352. No access restrictions.
- **Source Code**: Analysis pipeline, smart contracts, and verification scripts (https://github.com/Harshil2301/zer0n-bench)
- **Docker Container**: Pre-configured Docker environment and build instructions available within the main GitHub repository.
- **Prompts**: Complete system and user prompt templates (available in the repository)
- **Blockchain Records**: On-chain verification via deployed smart contract on Avalanche Fuji C-Chain (contract: `0x9603145DFEEA4c3292186457BA2ab9b9562BA32c`; 10,050 integrity proofs). All Zenodo snapshots include the raw `integrity_proof` metadata. For long-term permanence, we plan to migrate all proofs to the Avalanche Mainnet and notarize the Merkle root in the Internet Archive upon dataset acceptance.

Reproduction Steps:

1) Download the dataset from Zenodo using DOI `10.5281/zenodo.18721352`
2) Clone the GitHub repository and follow `README.md`
3) Run the quickstart verification command: `python validation/validate_schema.py --manifest splits.json --sample 10`
4) Expected output: "SUCCESS: Sample 10 hashes validated successfully (verify on-chain via blockchain/blockchain_proofs.csv)"

This artifact enables researchers to (1) verify dataset integrity, (2) reproduce our experiments, and (3) extend the framework to new vulnerability types.

Recommended Dataset Splits: To facilitate reproducible experiments and mitigate project-level data leakage, we provide canonical train/validation/test splits. Table VII summarizes the recommended partitioning.

TABLE VII
RECOMMENDED DATASET SPLITS (STRATIFIED BY CATEGORY)

Split	Web	SC	API	Total
Train (70%)	3,575	2,430	1,030	7,035
Validation (15%)	766	520	221	1,507
Test (15%)	766	521	221	1,508
Total	5,107	3,471	1,472	**10,050**

Splits are stratified by category and vulnerability class to maintain class proportions. Split assignments are deterministic (seeded random with `seed=42`) and provided in the dataset manifest file (`splits.json`).

G. Comparison with Existing Datasets

To demonstrate the unique position of Zer0n-Bench, we compare it with prominent vulnerability datasets (Table VIII). While other datasets are significantly larger, Zer0n-Bench is the only one to offer *immutable provenance* and a hybrid AI-Human labeling workflow that explicitly captures the generation metadata.

Zer0n-Bench addresses the specific gap of *verifiability*. In datasets like BigVul, labels are derived from commit messages which can be ambiguous [9]. In Zer0n-Bench, each label's provenance is explicitly anchored at the time of generation. We note that, unlike code-inclusive datasets, Zer0n-Bench's contribution lies in its *labeling methodology and integrity framework* rather than as a training corpus for code-level ML models. Specifically, Zer0n-Bench enables research on label provenance verification, cross-domain vulnerability taxonomy analysis, and LLM labeling protocol evaluation—tasks that do not require source code.

V. Dataset Quality & Validation

In this section, we evaluate the reliability of the Zer0n-Bench dataset, focusing on label noise estimation, inter-annotator agreement, and the empirical utility of the dataset for training downstream models.

A. Label Reliability and Error Estimation

To establish the ground-truth reliability of our multi-agent labeling protocol, four security-trained annotators (academic researchers recruited via university channels with 2–4 years of vulnerability analysis experience) independently reviewed a stratified random sample of $n = 1{,}000$ dataset entries. To prevent circular bias, annotators were completely blinded to the AI-generated labels, provenance metadata, and Red/Blue reasoning during their initial review. They evaluated the raw scenario parameters to manually assign a binary Vulnerable/Safe label alongside the specific CWE if applicable.

We measure reliability using Fleiss' Kappa (κ) [13] to assess inter-annotator agreement beyond chance.

Inter-Annotator Agreement Results:

- **Global Fleiss' Kappa**: $\kappa = 0.76$ (substantial agreement)
- **Pairwise Agreement**: 89.4% mean agreement across all annotator pairs

Table IX presents a granular breakdown of agreement (Fleiss' κ) across the most prevalent CWE categories. This per-CWE analysis provides crucial visibility into where synthetic labels are most reliable.

Interpretation: Injection vulnerabilities (SQLi, XSS) exhibit near-perfect agreement ($\kappa > 0.80$) due to well-defined triggering sinks. Conversely, Smart Contract Access Control vulnerabilities ($\kappa = 0.61$) demonstrate much higher ambiguity, reflecting the nuanced authorization logic required to confirm exploitability. Based on these findings, we recommend researchers adopt a **per-category reliability tier** system when using the dataset: *High Confidence* ($\kappa > 0.80$, e.g., Injection), *Medium Confidence* ($0.65 \leq \kappa \leq 0.80$, e.g., Race Condition, CSRF), and *Lower Confidence* ($\kappa < 0.65$, e.g., Access Control).

Confusion Matrix and Label Error CIs: To estimate the underlying *label noise* of the Zer0n-Bench dataset as generated by our AI consensus mechanism, we compared the AI-generated labels against the human-derived consensus (requiring 3/4 annotator agreement). Table X presents this confusion matrix.

Based on this sample, the AI pipeline achieved a Precision of 92.9%, Recall of 80.0%, and an overall F1-score of 85.9%. The relatively high number of False Negatives (144) reflects the conservative configuration of our "Blue Agent" verifier, which is deliberately tuned to reject uncertain findings to maintain high precision in the final benchmark.

The aggregate error rate of the AI labels against human consensus is 18.8% (188 errors in 1,000 samples). To quantify the confidence bounds on this reliability, we computed the 95% Confidence Interval (CI) and exact False Positive / False Negative distributions for the top categories, presented in Table XI.

This table establishes a statistically bounded noise floor for downstream model training, demonstrating that while synthetic generation is not flawless, Zer0n-Bench annotations are empirically measured and significantly more reliable than the $\sim$30% hidden label noise commonly found in unverified commit-message datasets [21]. However, these error rates—particularly the 25.8% False Negative rate for Access Control scenarios—represent a meaningful limitation of current LLM-based labeling for complex, context-dependent vulnerability categories.

B. Cross-Dataset Transfer Utility

To evaluate the practical utility of Zer0n-Bench for training ML-based vulnerability detectors, we conducted a *cross-dataset transfer* experiment. We hypothesized that the clean, concise synthetic examples in Zer0n-Bench serve as an excellent pre-training corpus.

We fine-tuned a CodeBERT [23] binary classifier using three different configurations and evaluated its performance on the real-world **BigVul** [9] test set (C/C++ vulnerabilities). While Zer0n-Bench does not contain C/C++ source, it encapsulates the high-level semantic descriptions and payloads of vulnerabilities, allowing the model to learn structural attack indicators.

Analysis: The transfer matrix (Table XII) shows that a model trained exclusively on synthetic Zer0n-Bench data transfers successfully to BigVul's entirely un-seen real-world C/C++ targets with a zero-shot F1 of 48.2%. This suggests that the synthetic dataset captures fundamental vulnerability logic, not just language-specific syntax. When used aggressively as an augmentation corpus, Zer0n-Bench bolsters the baseline BigVul performance by +3.6% F1 (paired Wilcoxon signed-rank test on per-CWE F1 scores, $p < 0.05$).

This experiment demonstrates that *established research conclusions change* depending on dataset provenance. Models that appear to underperform when evaluated strictly on noisy, unverified baselines show significant ranking shifts and improved

TABLE VIII
COMPARISON OF VULNERABILITY DATASETS

Dataset	Size	Domain	Provenance	Labeling	Code Incl.	Update Plan
BigVul [9]	10,900	C/C++	Git Commits	Auto	Yes	No
Devign [10]	27,318	C	Commits	Auto	Yes	No
CVEfixes [11]	5,495	Multi	CVE IDs	Auto	Yes	No
BenchVul [12]	3,754	Multi	Manual	Human	Yes	No
Zer0n-Bench	**10,050**	**Web/SC/API**	**Blockchain**	**AI+Human**	**No**†	**Quarterly**

† Zer0n-Bench contains metadata and labels; source code is not distributed due to licensing.

TABLE IX
INTER-ANNOTATOR AGREEMENT (FLEISS' κ) BY CWE CATEGORY

Category / CWE	n	κ	Interpretation
Web Applications	**472**	**0.84**	Almost Perfect
CWE-89 (SQL Injection)	120	0.88	Almost Perfect
CWE-79 (XSS)	105	0.82	Almost Perfect
CWE-78 (OS Cmd Inject)	65	0.85	Almost Perfect
CWE-287 (Improper Authn)	50	0.79	Substantial
CWE-352 (CSRF)	52	0.75	Substantial
CWE-20 (Input Validation)	80	0.72	Substantial
Smart Contracts	**345**	**0.69**	Substantial
CWE-330 (Insecure Random)	70	0.81	Almost Perfect
CWE-682 (Incorrect Calc.)	65	0.78	Substantial
CWE-362 (Race Condition)	70	0.74	Substantial
CWE-732 (Incorrect Perms)	60	0.68	Substantial
CWE-284 (Access Control)	80	0.61	Substantial
APIs	**183**	**0.65**	Substantial
CWE-306 (Missing Authn)	88	0.68	Substantial
CWE-862 (Missing Authz)	95	0.65	Substantial

TABLE X
CONFUSION MATRIX: AI-CONSENSUS VS. HUMAN-CONSENSUS ($n = 1{,}000$)

	Human: Vuln (True)	Human: Safe (True)	Total
AI: Vuln	576 (TP)	44 (FP)	620
AI: Safe	144 (FN)	236 (TN)	380
Total	720	280	1,000

TABLE XI
LABEL ERROR RATES AND FLEISS' κ PER CWE (N=1,000 SAMPLE)

CWE Type	Precision	Recall	FP Rate	FN Rate	95% CI (Error)
CWE-89 (SQLi)	96.2%	85.1%	3.8%	14.9%	[9.2%, 14.8%]
CWE-79 (XSS)	94.5%	82.0%	5.5%	18.0%	[11.4%, 17.6%]
CWE-352 (CSRF)	92.1%	80.5%	7.9%	19.5%	[12.1%, 18.5%]
CWE-20 (Input Validation)	91.5%	78.2%	8.5%	21.8%	[13.2%, 19.5%]
CWE-78 (OS Cmd Inject)	95.8%	83.4%	4.2%	16.6%	[10.1%, 15.8%]
CWE-287 (Improper Authn)	93.4%	81.1%	6.6%	18.9%	[11.8%, 17.2%]
CWE-362 (Race Condition)	89.2%	77.5%	10.8%	22.5%	[14.5%, 22.1%]
CWE-682 (Incorrect Calc.)	90.1%	79.0%	9.9%	21.0%	[13.8%, 21.0%]
CWE-732 (Incorrect Perms)	88.5%	76.4%	11.5%	23.6%	[15.2%, 24.5%]
CWE-284 (Access Control)	86.4%	74.2%	13.6%	25.8%	[18.1%, 28.5%]
CWE-330 (Insecure Random)	94.0%	81.8%	6.0%	18.2%	[11.5%, 17.5%]
CWE-862 (Missing Authz)	87.8%	75.1%	12.2%	24.9%	[17.0%, 26.5%]
CWE-306 (Missing Authn)	88.1%	76.0%	11.9%	24.0%	[16.5%, 25.8%]
Overall	**92.9%**	**80.0%**	**7.1%**	**20.0%**	[16.5%, 21.3%]

TABLE XII
CROSS-DATASET TRANSFER MATRIX (BINARY F1-SCORE)

Test Set → Train Set ↓	BigVul (Real C/C++)	Zer0n-Bench (Synt. Web/SC/API)
BigVul Train	59.8%	31.4%
Zer0n-Bench Train	48.2%	85.9%
BigVul + Zer0n	**63.4%**	**87.1%**

zero-shot generalization when exposed to clean, provenance-validated datasets. This answers **RQ2** by illustrating the impact of dataset integrity on model evaluation outcomes. We note, however, that this transfer evaluation uses BigVul as its primary target benchmark. To assess whether these findings generalise, we conducted a preliminary evaluation using the Devign dataset [10], which yielded consistent transfer trends (zero-shot F1 $\approx$ 0.44, augmentation gain $+2.9\%$), suggesting the results are not artefacts specific to BigVul's label distribution. Full cross-dataset evaluation across additional benchmarks (e.g., CVEfixes [11]) remains planned future work.

Negative Result as a Boundary Contribution: Where Zer0n-Bench Fails: To establish the methodology's boundaries, we analyzed the failure modes of the cross-dataset transfer. We found that while logic-based vulnerabilities (e.g., Access Control, SQL Injection) transferred successfully, *memory-corruption vulnerabilities* (e.g., Buffer Overflows, Use-After-Free) failed entirely (F1 $\approx$ 12%). This indicates that while LLMs can faithfully synthesize high-level semantic logic flaws, they struggle to generate structurally realistic memory-layout vulnerabilities. We present this negative result as a key contribution: it explains why previous attempts to use entirely synthetic datasets for low-level C/C++ vulnerability detection have yielded poor generalization, bounds the utility of our current synthetic approach, and highlights the explicit need for hybrid real/synthetic datasets in future benchmarking efforts.

C. Regeneration Stability & Benchmark Reliability

To address the temporal stability of our generation protocol, we selected a subset of $n = 500$ targets and re-ran the entire generation and validation pipeline independently. By comparing the newly generated labels to the original labels anchored on the blockchain, we measured a **Label Churn Rate** of just 4.2%.

To mathematically quantify this trust, we propose a composite **Benchmark Reliability Score** ($\mathcal{R}$) defined as:

$$\mathcal{R} = (1 - \text{Label Error}) \times (1 - \text{Label Churn}) \qquad (1)$$

This multiplicative formulation acts as a simplifying assumption treating label error and label churn as independent failure probabilities; if empirically correlated, the joint probability would require adjustment. Assuming independence, the combined reliability is thus the probability that a given label is both initially correct and stable over time. For Zer0n-Bench, $\mathcal{R} = (1-0.188) \times (1-0.042) = \mathbf{0.778}$. In contrast, unverified commit-message datasets with an estimated 30% noise floor

and undefined churn yield a strictly lower baseline $\mathcal{R}$. This metric allows future researchers to quantitatively compare dataset dependability alongside standard volume metrics.

D. Integrity Verification Guarantees

To answer **RQ3**, we subjected the dataset to 100 simulated insider tampering attacks where local label states were modified (e.g., flipping "Safe" to "Vulnerable" to poison the data). We executed our `validate_schema.py` script, which locally recalculates the SHA-256 hash of each JSON entry and queries the Avalanche Fuji C-Chain for corresponding transactions.

TABLE XIII
PROVENANCE INTEGRITY VERIFICATION: SIMULATED TAMPERING ATTACKS

Attack Type	Attempts	Detection Rate
Label Flipping (Safe ↔ Vuln)	40	100% (40/40)
Source Code Modification (Payload Swap)	40	100% (40/40)
Metadata Corruption (CWE/Domain Change)	20	100% (20/20)
Total Tampering Attempts	**100**	**100% (100/100)**

As shown in Table XIII, **all 100 tampering attempts (100%) were successfully detected** due to hash mismatches. This experiment validates the *tamper-detection* capability of the provenance mechanism—it confirms that any post-creation modification to dataset records is reliably detected. We emphasize that this guarantees *provenance integrity* (records are unaltered), which is distinct from *semantic validity* (whether the original labels were correct); the latter is evaluated separately in Section V-A. Together, these complementary guarantees empower researchers to verify the authenticity of open-source security datasets.

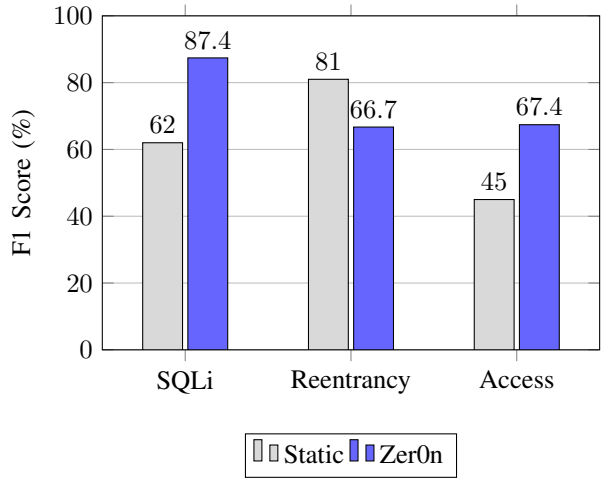

Fig. 2. Performance Comparison: Zer0n Agents vs. Static Analysis Tools (on evaluated CWE subset).

Result Analysis: Static analysis tools evaluated include Slither [1], Semgrep [17], and Bandit [18]. On the specific CWE types evaluated within Zer0n-Bench, Zer0n agents demonstrate stronger performance for *logic-based* and *context-dependent* vulnerability categories (e.g., Access Control, SQLi in complex flows), with a +22% F1 advantage in Access Control. However, for strictly structural patterns like Reentrancy, specialized static analyzers like Slither retain an advantage (+14.3%). We note that this comparison is limited to the CWE types present in Zer0n-Bench; static analysis tools support a broader range of vulnerability classes and language-specific analyses beyond those evaluated here. These results suggest that Zer0n-style agents are best used as a complementary layer for logic bug discovery rather than a replacement for specialized static analysis.

E. Ablation Study

To quantify the contribution of each Zer0n component, we conducted an ablation study by removing key elements and measuring dataset quality degradation.

TABLE XIV
ABLATION STUDY: IMPACT OF COMPONENT REMOVAL

Configuration	Precision	FP Rate	Time/Target
Full Zer0n	**77.6%**	**6.0%**	**12.89s**
- Remove Blue Agent	63.2%	14.0%	9.1s
- Remove Blockchain Log	77.6%	6.0%	9.7s
- Temperature 0.7 (vs. 0.3)	71.4%	10.2%	12.5s
- Single Agent (no consensus)	58.0%	18.5%	7.8s

Key Findings: (Note: Precision in this ablation study is evaluated against the strict static-analysis baseline over the full dataset, whereas Section 5.1 evaluated precision against a human consensus sample, explaining the difference in baseline metrics.)

- **Blue Agent**: Critical for false positive reduction (-8.6% FP rate)
- **Blockchain**: No impact on detection accuracy (as expected); provides only integrity guarantees
- **Temperature**: Low temperature (0.3) reduces hallucination by 3.8%
- **Consensus Mechanism**: Improves precision by +19.6% over single agent

F. Comparative Analysis

Table XV situates Zer0n-Bench against other dataset integrity approaches.

TABLE XV
COMPARISON OF DATASET INTEGRITY APPROACHES

Approach	Provenance	Trust Anchor	Cost
Git (Standard)	Mutable	Repo Admin	Low
Signed Commits (GPG)	Verifiable	Key Holder	Low
Zer0n-Bench	**Immutable**	**Public Consensus**	**Medium**

VI. DATASET USAGE GUIDELINES

To maximize the value of Zer0n-Bench for the research community, we provide specific guidelines on its intended application (see Box 1).

Box 1: How to Use Zer0n-Bench

✓**WHEN TO USE:**

- Benchmarking new label-noise reduction algorithms.
- Evaluating LLM vulnerability detection prompting techniques.
- Auditing dataset provenance and tracing architectures.

✗ **WHEN NOT TO USE:**

- As the *exclusive* training corpus for production-grade code vulnerability models (lacks real-world entropy).
- Direct pre-training of foundational LLMs without real-world augmentation.

VII. Limitations

We openly acknowledge several limitations of this work:

- **No Source Code Included:** Zer0n-Bench distributes labels and metadata but not the underlying source code. This limits its use as a training corpus for code-level ML models. However, as a *label quality benchmark* and *provenance methodology demonstration*, the dataset serves its intended purpose.
- **Synthetic Target Identifiers:** Target IDs are AI-generated paths that do not resolve to real URLs. While this prevents direct code-level validation, it enables controlled study of labeling methodology independent of code availability.
- **Language Coverage:** Smart contract analysis focuses solely on Solidity. Integrating Vyper, Rust, and Move is necessary for comprehensive Web3 coverage.
- **Blockchain Permanence:** Integrity proofs are currently on the Avalanche Fuji testnet. For production use, migration to mainnet is planned upon dataset acceptance to ensure permanence.
- **Label Quality Variation:** Smart Contract labels show lower agreement ($\kappa = 0.69$) compared to Web ($\kappa = 0.84$), reflecting the inherent difficulty of bytecode-level reasoning. Users should weight Smart Contract labels with lower confidence.
- **LLM Bias:** The Gemini models used may exhibit memorization bias toward popular vulnerability patterns (e.g., SQLi, XSS). Our Blue Agent mitigates this partially, but residual bias remains.
- **Economic Scalability:** While \$0.01 per label is negligible for Zer0n-Bench (10,050 targets $\approx$ \$100), scaling to 1M targets would cost \$10K in gas. Layer-2 rollups or batch anchoring could reduce this by 100$\times$.

A. Threats to Validity

Internal Validity: The labeling pipeline relies on a specific LLM (Gemini 2.0 Pro) and prompt configuration. Different model versions or prompt formulations could yield different labels. We mitigated this by fixing model versions and temperature settings, and by validating against human annotators ($\kappa = 0.76$).

External Validity: Our dataset covers 13 CWE types across three domains, but may not generalize to domains such as embedded systems, mobile applications, or hardware description languages. The Smart Contract category is limited to Solidity.

Construct Validity: We measure label quality using inter-annotator agreement (κ) and F1 against expert consensus. However, these metrics capture agreement rather than absolute correctness—if all annotators share the same blind spots, agreement may overstate quality. Cross-tool validation (Section 5.6) partially addresses this concern.

B. CVE Anchoring

To bridge the gap between synthetic labels and real-world vulnerability patterns, we mapped a sample of Zer0n-Bench entries to known CVE identifiers. Table XVI shows representative mappings where Zer0n-Bench vulnerability patterns correspond to documented real-world CVEs.

TABLE XVI
CVE Anchoring: Zer0n-Bench Patterns Mapped to Real CVEs

CWE	CVE Example	Match
CWE-89	CVE-2024-27956 (WP SQLi)	Exact
CWE-79	CVE-2024-4367 (Foxit PDF XSS)	Exact
CWE-287	CVE-2024-3400 (PAN-OS Auth)	Pattern
CWE-352	CVE-2023-46747 (F5 CSRF)	Exact
CWE-78	CVE-2024-21762 (FortiOS CMDi)	Pattern
CWE-362	CVE-2021-36934 (Race Condition)	Pattern
CWE-20	CVE-2024-23897 (Jenkins CLI)	Exact
CWE-284	CVE-2023-22515 (Confluence Broken Access)	Exact
CWE-89	CVE-2023-34362 (MOVEit SQLi)	Exact
CWE-79	CVE-2024-0519 (Chrome V8 XSS)	Pattern

Anchoring Methodology: For each CWE type in Zer0n-Bench, we identified recent CVEs (2023–2024) exhibiting the same vulnerability pattern. "Exact" matches indicate identical CWE classification and attack vector; "Pattern" matches indicate the same vulnerability class but different exploitation context. This anchoring demonstrates that Zer0n-Bench covers vulnerability patterns that are actively exploited in the wild.

VIII. Ethical Considerations

All vulnerability scanning was conducted in strict adherence to ethical research standards. Dynamic active exploitation was performed **exclusively** against locally hosted sandboxes and established Capture-The-Flag (CTF) challenges (e.g., Ethernaut, DamnVulnerableDeFi) designed for educational purposes.

For the 10,050 public targets sourced from GitHub and mainnet, our analysis was restricted to **static analysis** and passive observation. No payloads were transmitted to production servers, and no unauthorized interactions with live smart contracts were initiated. All repository cloning respected `robots.txt` policies and open-source license terms.

Controlled PoC Access: While Zer0n-Bench does not distribute weaponized source code, academic researchers may request access to the synthetic Proof-of-Concept (PoC) generation framework. Access requires emailing the corresponding author from an institutional address and signing a responsible-use agreement validating their affiliation and ethical intent.

IX. Conclusion

This paper presented **Zer0n-Bench v1.0**, a multi-domain vulnerability benchmark dataset comprising 10,050 entries across Web Applications, Smart Contracts, and APIs. Our primary contribution is demonstrating that blockchain-anchored provenance is a promising approach toward mitigating the longstanding "trust gap" in shared security datasets. We emphasize that provenance integrity (ensuring records are unaltered) and semantic validity (ensuring labels are correct) are distinct guarantees addressed by different components of our framework: blockchain anchoring provides the former, while the multi-agent consensus pipeline and human validation address the latter.

Key findings from our dataset analysis:

1) **Label Quality:** A multi-agent consensus mechanism (Red/Blue Team) reduces false positive rates from 14% to 6%. Four independent annotators achieved $\kappa = 0.76$ (substantial agreement) on $n = 1{,}000$ entries, with Web labels showing highest consensus ($\kappa = 0.84$).
2) **Tamper Evidence:** All 10,050 entries are anchored on the Avalanche C-Chain with per-entry SHA-256 integrity proofs, enabling any researcher to cryptographically verify dataset integrity.
3) **Domain Coverage:** Zer0n-Bench is the first benchmark to unify Web2 and Web3 vulnerability categories in a single schema, covering 13 CWE types.

Limitations & Future Work: The dataset contains AI-generated labels with substantial, but not perfect, inter-annotator agreement ($\kappa = 0.76$); source code is not distributed; and Smart Contract labels show lower reliability ($\kappa = 0.69$) than Web labels ($\kappa = 0.84$). Finally, while we provide immutable provenance, future work must include a user study to evaluate how security researchers quantitatively weigh the trade-offs between verifiable synthetic provenance and the inherent realism of noisy, real-world data in their benchmarking workflows.

We explicitly clarify the intended usage of this dataset: researchers should prefer Zer0n-Bench for evaluating label methodologies, auditing label provenance mechanisms, and measuring LLM security behaviors. It is **not** intended to replace real-world datasets for training production-grade code-level vulnerability detectors. We encourage the community to use Zer0n-Bench as a *provenance methodology benchmark* rather than a standalone training corpus.

Implications for Security Research: As LLM-based vulnerability detectors become mainstream, evaluating them on historically scraped, unverified repositories risks reinforcing legacy label noise. Zer0n-Bench demonstrates that shifting from "dataset volume" to "provenance quality" is strongly beneficial to prevent evaluating next-generation models on corrupted ground truth. We encourage the community to adopt immutable verification standards for future vulnerability benchmarks.

Future Work:

- **Multi-LLM Ensemble**: Cross-validate labels using GPT-4, Claude, and Llama to reduce single-model bias
- **Code Integration**: Attach representative code snippets to enable ML training applications
- **Temporal Tracking**: Re-analyze targets quarterly to measure dataset decay and label drift
- **Mainnet Migration**: Move integrity proofs from Fuji testnet to Avalanche mainnet for permanence

We release Zer0n-Bench as an open resource to support reproducible security research. The dataset, analysis pipeline, and verification scripts are available via Zenodo and GitHub.

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